Fatima Sabiu Maikore, Ph.D.

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Nigeria (Open to Global Remote Roles)

Professional Summary

Data & Ontology Specialist with 10+ years of experience designing and delivering complex data systems across healthcare, research, and enterprise environments. Combines data modelling expertise (ERD, relational & NoSQL) with semantic technologies and LLM-driven alignment to standardise data, translate complex requirements into scalable architectures, standards, validation rules, and reporting solutions, and support decision-making.

Led production systems in high-pressure environments, including healthcare platforms supporting over 450 users, and developed ontology-driven tools for large-scale semantic integration.

Core Expertise

- **Data Modelling & Architecture:** ERwin, MySQL Workbench, Oracle, MySQL, SQL Server, MongoDB schema design, multi-module systems
- **Semantic Technologies:** RDF, OWL, SHACL, SPARQL, knowledge graphs, LLM-based ontology alignment, IEEE 1872.1
- **Data Governance:** Data dictionaries, validation rules, naming standards, schema documentation
- **Analytics:** SQL optimisation, Power BI, custom dashboards (HTML, CSS, JavaScript)
- **Programming:** Python, SQL, Java, JavaScript, Node.js

Professional Experience

2024 – Present

Lead Developer — Electronic Medical Records System

Royal Specialist Hospital, Abuja

- Designed and implemented a production EMR system supporting hospital operations
- Developed multi-module architecture: patient management, consultations, lab, pharmacy, billing, inpatient/outpatient
- Built database schema from scratch (relational + MongoDB)
- Defined data standards, validation rules, and system documentation
- Developed dashboards for operational monitoring and reporting

Impact:

- Improved efficiency across clinical workflows
- Increased transparency in billing systems
- Strengthened drug inventory management and tracking

Professional Experience (continued)

2017 – 2023

Support Analyst — Electronic Medical Records System
Nigerian Hajj Medical Team

- Supported EMR system used by approximately 450 healthcare professionals during Hajj operations
- Optimised reporting pipelines for disease surveillance and public health analytics
- Enabled system performance in a high-pressure, time-bound environment (seasonal deployment)

Impact:

- Enabled mandatory public health reporting to Saudi authorities
- Supported disease tracking and outbreak monitoring

2021 – 2024

Research Associate / University Teacher
University of Sheffield

- Designed ontology-driven systems for healthcare and behavioural science applications
- Developed semantic tools and frameworks for data integration and knowledge representation
- Delivered teaching in databases, semantic web, and software engineering
- Also served as Deputy Director, Centre for Inclusive Computing, leading onboarding and ambassador schemes and widening participation through national inclusion initiatives.

2013 – 2021

Lecturer
Baze University

- Taught databases, semantic web, and software engineering modules
- Supervised 50+ undergraduate projects involving real-world system design
- Led curriculum improvements aligned with industry standards

Selected Projects

LLM Ontology Alignment Tool

Developed a large language model (LLM)-driven system for ontology alignment and semantic integration across domains. Built user interface for uploading, analysing, and exporting ontology matches. Enabled scalable interoperability for research and applied systems.

ELECTRICA Ontology

Designed clinical ontology integrating SNOMED-CT, ICD, and BFO. Published on NCBO BioPortal to support knowledge sharing and interoperability in healthcare.

TURRBO Platform

Developed the Relationship Between Behaviours Ontology (RBBO) for TURRBO (Tools for Understanding Relationships Between Behaviours Using Ontologies), with RBBO accepted into and hosted by the Behavioural and Social Sciences Ontology (BSSO) Foundry and built from reused Behaviour Change Intervention Ontology (BCIO) modules. Created ontology-driven tools for behavioural science and managed the supporting infrastructure across an internal Ubuntu server and AWS-backed services for semantic integration, cross-domain reasoning, scalability, and remote access.

Selected Projects (continued)

IEEE 1872.1 Standard

- Developed OWL ontology for robot task representation. Collaborated with international working group to ensure semantic alignment and interoperability.

Education

2014 – 2020
- **Ph.D., Computer Science. Goldsmiths, University of London**
  Thesis title: *A semantic-driven framework for IT support of clinical laboratory standards.*

2009 – 2010
- **M.Sc. Information Systems: Business IT. The University of Manchester**

2005 – 2009
- **B.Sc. Information Systems. American University of Nigeria**

Certifications

2020
- **Cloud Application Developer**, IBM Skills Academy.

2019
- **Database Programming with PL/SQL**, Oracle Academy.
- **Database Design and Programming with SQL**, Oracle Academy.
- **Java Programming**, Oracle Academy.

2009
- **ERP Solution Architect**, SAP Global Certification.

Professional Memberships

2024
- **Fellow of the Higher Education Academy (FHEA)**

2019
- **Member of the Institute of Electrical and Electronics Engineers (IEEE)**
- **Member of the Association for Computing Machinery (ACM)**

2017
- **Google Women Techmakers Ambassador**
- **Member of the Organization for Women in Science for the Developing World (OWSD)**

2017 - present
- **Robot Task Representation Working Group** , IEEE Standards Association.

References

Available on Request